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HTML responsive design 30m

Schade (2014)^{fnref:b09b73} defines responsive web design as follows:

Responsive web design (RWD) is a web development approach that creates **dynamic changes** to the appearance of a website, **depending on the screen size and orientation** of the device being used to view it. RWD is one approach to the problem of designing for the multitude of devices available to customers, ranging from tiny phones to huge desktop monitors.

We will now discuss the HTML options for supporting responsive web design.

HTML viewport

A viewport is the area in which a user can see the document. We can set up the viewport using the name and content attributes in the `<meta>` element. Typically, a responsive website should include the following tag:

```
<meta name='viewport' content='width=device-width, initial-scale=1.0'>
```

It defines 100 per cent of the viewport with an initial zoom level of 1 (default value).

Responsive images

To optimise images for mobile and desktop users, we can use HTML5's `<picture>` tag.

Using the `<picture>` tag gives you more flexibility in creating the image resource. This ensures you have a responsive image.

When using a `<picture>` tag, you will use the `<source>` element to specify the image resource and the `srcset` attribute to specify the URL of the image.

View the CodePen example below to see how the `<picture>` element is used to create responsive images.



HTML

Result

EDIT ON

LIVE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width,
initial-scale=1.0">
  <title>Responsive Images
Example</title>
</head>
<body>
  <h1>HTML Responsive Images -
Example</h1>
  <picture>
    <source media="(min-width:
700px)"
srcset="https://media.geeksforgeeks.
org/wp-
content/uploads/20190825000042/geeks
-221.png">
    <source media="(min-width:
450px)"
srcset="https://media.geeksforgeeks.
org/wp-
content/uploads/20190802021607/geeks
14.png">
    fnref:704025</sup> to learn more about Cloudinary.

Next, you'll be tasked to create an image series using responsive design principles.

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This task will be valuable for your assessments.

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☒ Designing a responsive image series

Non-Assessed

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**Status: Done**

Reset ↺

Using [Cloudinary](#), take the large [Cactus flower.jpg](#) image (3000x4000 pixels) and create five images in different sizes for responsive web content. The smallest image should be 240 pixels wide, and the largest should be 1024 pixels wide.

☒ I have completed the task

The following settings should have been selected.

**Breakpoints Generation Settings:**

- **Resolution range:** from 200 to 1024 pixels
- **Size step:** 20 KB
- **Maximum images:** 20 images
- **Retina resolution:** include double resolution (DPR 2.0) images

This will generate the following image sizes:

| Image | Width | Height | File size |                   |
|-------|-------|--------|-----------|-------------------|
| 1     | 200   | 267    | 9.3 KB    | <b>View image</b> |
| 2     | 459   | 612    | 30.8 KB   | <b>View image</b> |
| 3     | 663   | 884    | 50.8 KB   | <b>View image</b> |
| 4     | 828   | 1104   | 68.7 KB   | <b>View image</b> |
| 5     | 990   | 1320   | 88.8 KB   | <b>View image</b> |
| 6     | 1024  | 1365   | 94.1 KB   | <b>View image</b> |

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The images can be used in HTML as follows:

```

```

Previous attempts ▼



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Thoughts on this page?

